8. Wednesday, April 16. Number system Axioms. Some proof Techniques (3.1, 3.2)

Field Axioms (properties of addition and multiplication)

1) Closure

2) Commutativity

3) Associativity

4) Distributivity

5) Existence of identities: There are two different real numbers 0 and 1 such that for every $a \in \mathbb{R}$ we have $a+0=a$ and $a \cdot 1 = a$.

6) Additive inverses: For every $a \in \mathbb{R}$, there exists $-a \in \mathbb{R}$ such that $a + (-a) = 0$

7) Multiplicative inverses: For every $a \in \mathbb{R}$ s.t $a \neq 0$, there exists $a^{-1} \in \mathbb{R}$ such that $a \cdot a^{-1} = 1$

Notation: For $a, b \in \mathbb{R}$ we denote $a - b = a + (-b)$

If $b \neq 0$, we denote $\frac{a}{b} = ab^{-1}$

Remark:

1) The naturals $\mathbb{N}$ satisfy axioms (1)–(5)
2) The integers $\mathbb{Z}$ satisfy axioms (1)–(6)
3) The rationals $\mathbb{Q} = \left\{ \frac{a}{b} \mid a, b \in \mathbb{Z},\ b \neq 0 \right\}$ satisfy (1)–(7)

Equality Axioms: (For any notion of equality)

8) Reflexivity: For every $a \in \mathbb{R}$, we have $a = a$.

9) Symmetry: For any $a, b \in \mathbb{R}$, if $a = b$ Then $b = a$.

10) Transitivity: For all $a, b, c \in \mathbb{R}$, if $a = b$ and $b = c$, Then $a = c$.

11) Substitution: For all $a, b \in \mathbb{R}$, if $a = b$ Then b may be substituted for a in any mathematical statement without changing The Truth value of That statement.

AT This point, we may only assume definitions and axioms 1–11 for our proofs!

Theorem 1: Let $a, b, c \in \mathbb{R}$. If $\overbrace{a = b}^{P}$, Then $\overbrace{ac = bc}^{Q}$ and $a + c = b + c$.

Proof: We fix $a, b, c \in \mathbb{R}$ and suppose That $a = b$. We prove $ac = bc$ first.

By closure of $\mathbb{R}$, we know That $ac \in \mathbb{R}$

By reflexivity of equality $ac = ac$.

Then, by the substitution axiom, we may substitute b for a on The right-hand side of The equality To deduce That $ac = bc$, as desired.

Similarly, by closure, $a + c \in \mathbb{R}$. Now since $a = b$, by substitution we get $a + c = b + c$

Hence $ac = bc$ and $a + c = b + c$, as desired. □

- Complete English sentences, properly punctuated.
- Each step is justified and follows order and direction.
- Various versions of □ To flag end of proof.

Theorem 2: If $a,b,c \in \mathbb{R}$ are such that $a+c=b+c$, then $a=b$.

Seems obvious but needs a proof. Not true for multiplication!

Proof: Suppose a,b,c are such that $a+c=b+c$. By existence of additive inverse, we have $-c \in \mathbb{R}$ such that $c+(-c)=0$. Now, since $a+c=b+c$, by our previous Theorem, we have

$$(*)\quad (a+c)+(-c) = (b+c)+(-c)$$

Using associativity, we get

$$(a+c)+(-c) = a+(c+(-c)) \quad \text{and} \quad (b+c)+(-c) = b+(c+(-c))$$

So we can substitute in both sides of $(*)$ to get

$$a+(c+(-c)) = b+(c+(-c))$$

Since $c+(-c)=0$, we can substitute in both sides to get

$$a+0 = b+0$$

But, by the additive identity axiom, $a+0=a$ and $b+0=b$. So by substitution,

$$a = b.$$

As desired $\square$

Technique (Proving Universal Statements)

To prove a statement of the form "for all $a \in A$, we have $P(a)$"

We fix an arbitrary element $a \in A$ and prove $P(a)$ holds

Can only use $a \in A$.

Example: For all integers x, if x is even then x^2 is even.

Not a proof: Let 8 be an even integer. Then $8^2 = 64 = 2 \cdot 32$ is even.

Theorem 3: For any real number a, we have $0 + a = a$.

Proof: Fix a real number a. By the "existence of identities" axiom, we know that $a + 0 = a$. By commutativity, we also know that

$$0 + a = a + 0.$$

Since $0 + a = a + 0$ and $a + 0 = a$, by the Transitivity axiom we get $0 + a = a$, as desired $\square$

Theorem 4: In the real numbers, the additive identity is unique.
(There is only one real number with the property $a+0=a$ for every $a \in \mathbb{R}$)

Technique: (Proving uniqueness) In order to prove something is unique,
1) suppose you have two of that thing, say x and y
2) Show $x=y$.

Proof of Theorem 4: Suppose $0' \in \mathbb{R}$ is another additive identity, so $b+0'=b$ for any $b \in \mathbb{R}$

Since $0'$ is an additive identity, we get $0+0'=0$.

Since 0 is an additive identity, we get $0'+0=0'$.

But by commutativity, $0+0' = 0'+0$.

So by substitution $0' = 0$, as desired □

Another proof: Assume $a+0=a$ and $a+0'=a$ for any $a \in \mathbb{R}$.
By symmetry, we have $a = a+0'$. Since $a+0=a$ and $a = a+0'$ by transitivity we have $a+0 = a+0'$. By cancellation, (7.2) we get $0=0'$ □